

SYMBOLS		
$\bar{x}$ "x-bar" your sample's average	μ "mew" true population mean, unknown	μ_0 the claimed number — the one in H₀
s sample SD, from your data	n sample size $\sqrt{n}$: $\sqrt{25} = 5$	SE standard error = $\sigma/\sqrt{n}$ or $s/\sqrt{n}$
β "beta" Type II rate; Power = $1-\beta$	df 1 mean $n-1$ · slope $n-2$ · MLR $n-k-1$	H₀ the null — always has an =
$\hat{y}$ "y-hat" a predicted y	e residual = actual – predicted	b₀ b₁ intercept / slope from your data
r correlation, –1 to +1	R² % of variation in y explained	RMSE typical prediction error
z .05 → 1.645 · .025 → 1.96	z .01 → 2.326 · .005 → 2.576	t* critical t — like z but needs df
		σ "sigma" population SD, only if given
		α "alpha" the bar you compare p to
		H₁ the alternative: <, > or $\neq$
		β_0 β_1 the true ones; H ₀ : $\beta_1 = 0$
		k how many predictors
		Σ add it up over every data point

Q1 · ONE-SAMPLE MEAN — Analyze > Distribution

Before part a: is it t or z?

RULE only **s** given → **t**, df = n-1 · **σ** given, or power is asked → **z**

Raw column of numbers → Distribution > Test Mean. Just n, $\bar{x}$, s in a sentence or table → TestMean.jsl.

a Is the variable categorical or quantitative?

JMP none – read the problem

SAY "The _____ of each _____. **Quantitative** — a measured number that makes sense to average."

b Define the parameter in context

JMP none

SAY " μ = the true mean _____, in _____, for all _____."

Lose the point by: writing $\bar{x}$, describing the sample, or dropping the units / population.

c State the null and alternative

JMP none – use the wording table below

SAY H₀: $\mu =$ _____ vs H₁: μ [**<** / **>** / **$\neq$**] _____

The same claimed number goes in both lines. H₀ always gets the "=".

WORDS IN THE PROBLEM	H ₁	READ THIS LINE	CRIT .05/.01
more than, greater, exceeds, increased, above	$\mu > \mu_0$	Prob > t	1.645 / 2.326
less, reduced, decreased, improved (faster), below	$\mu < \mu_0$	Prob < t	–1.645 / –2.326
different from, changed, not equal, "off", "is the claim right?"	$\mu \neq \mu_0$	Prob > t	±1.96 / ±2.576

Direction comes from the **claim being tested**, not from background motivation in the story.

d Check the conditions

JMP Analyze > Distribution > ▼variable > Normal Quantile Plot · then Continuous Fit > Normal > ▼Fitted Normal > Goodness of Fit

SAY Independence: the _____ are a random sample and fewer than 10% of all _____. **Normality:** the problem states bell-shaped, **or** n = _____ ≥ 30 so the CLT applies, **or** the dots follow the straight line and Shapiro-Wilk Prob<W = _____ > α .

e Find the test statistic and p-value

JMP Analyze > Distribution > Y = variable > OK > ▼variable > Test Mean > type μ_0 > leave the SD box empty for t, or fill σ for z > OK

OR Help > Sample Index > Tool > Calculators > TestMean.jsl – takes n, $\bar{x}$, s with no raw data

SAY SE = $s/\sqrt{n} =$ _____ / $\sqrt{}$ _____ = _____ ; t = $(\bar{x} - \mu_0)/SE =$ _____ , df = _____ , p = _____

The *Test Mean* panel prints all three tails. Match the one to H₁ — the biggest lost point on the whole exam. A one-tailed p is exactly half the two-tailed one; if yours isn't, one of the two is wrong.

f Interpret the test statistic

SAY "The sample mean sits _____ standard errors [**above / below**] the claimed _____."

t is a *distance* measured in standard errors. |t| < 1 ordinary · ≈ 2 borderline · > 3 rare.

g Interpret the p-value

SAY "If μ really were _____, there is a _____ % chance of getting a sample mean at least this [**high / low / far**] from _____ by random chance alone."

Never mention α and never say "reject" here — that belongs in the conclusion.

h State the conclusion

SAY "Since p = _____ [**<** / **>**] $\alpha =$ _____, we [**reject / fail to reject**] H₀. There [**is / is not**] enough evidence to say that _____."

Never "accept H₀" and never "prove". If your instructor words it "sufficient evidence to conclude", that's the same sentence.

i Confidence interval for the mean

JMP Summary Statistics already lists Upper / Lower 95% Mean · other levels: ▼variable > Confidence Interval > 0.90 / 0.99

SAY $\bar{x} \pm t^*SE =$ _____ ± _____ → "We are _____ % confident the true mean _____ for _____ is between _____ and _____."

Cross-check: if the interval **contains** μ_0 you will fail to reject; if it **excludes** μ_0 you will reject.

When the problem gives only summary numbers

SAY If they hand you s^2 (a **variance**), square-root it first. Home Prices: $s = \sqrt{7,317,400,000} \approx 85,542$; SE = $85,542/\sqrt{69} \approx 10,298$; t = $(141,400 - 155,000)/10,298 \approx -1.32$

"The sample statistic is 1.03 standard errors below the hypothesised value"

SAY That sentence **is** the answer: t = –1.03. Below → negative, above → positive. df = n – 1.

READING A JMP DISTRIBUTION REPORT

PANEL ROW	IS REALLY
Mean	$\bar{x}$
Std Dev	s — the sample SD
Std Err Mean	SE — already computed, don't redo $s/\sqrt{n}$
Upper / Lower 95% Mean	a ready-made 95% CI for μ
N	n
Quantiles	min, Q1, median, Q3, max — use to judge skew

Q2 · POWER & TYPE I / II ERROR — mostly by formula, not by menu

The whole idea in 3 steps — works for any power question

1 SE = $\sigma/\sqrt{n}$

2 cut-off on the $\bar{x}$ scale, centred at μ_0 : right $\mu_0 + z\alpha SE$ · left $\mu_0 - z\alpha SE$ · two-sided $\mu_0 \pm z\alpha/2 SE$ (two cut-offs)

3 re-centre on the **true** μ : $z = (\text{cut-off} - \mu_{\text{true}})/SE$, then read the reject-side area = **Power**, and $\beta = 1 - \text{Power}$

The cut-off never moves — only the centre of the curve moves.

a Describe the sampling distribution of $\bar{x}$ under H₀

JMP none – it is just μ_0 and $\sigma/\sqrt{n}$

SAY Shape: approximately normal, because _____. **Mean:** $\mu_0 =$ _____. **Standard deviation:** $\sigma/\sqrt{n} =$ _____ / $\sqrt{}$ _____ = _____.

Reason blank: " σ is known and the population is stated normal" or " $n \geq 30$, so the CLT applies".

b For what values of $\bar{x}$ is H₀ rejected?

SAY cut-off = $\mu_0 + (z)(SE) =$ _____ + (_____)(_____) = _____ → "reject H₀ for any sample mean [**above / below**] _____."

z carries the sign of the tail: –1.645 for a "less than" test, +1.645 for "greater than", ±1.96 two-sided at $\alpha = .05$.

c Power, and the Type II rate β

SAY $z = (\text{cut-off} - \mu_{\text{true}})/SE = ($ _____ $-$ _____ $)/$ _____ $=$ _____ ; Power = _____ and $\beta = 1 - \text{Power} =$ _____

Two-sided: work out **both** tails and add them (one is usually tiny, but show it).

Worked example — Soap

$\mu_0 = 33$, $\sigma = 12.4$, $n = 35$, $\alpha = .05$ "less than", true mean 29 SE = $\sigma/\sqrt{n} = 12.4/\sqrt{35} = 12.4/5.9161 \approx 2.096$ cut-off = $33 + (-1.645)(2.096) = 33 - 3.448 \approx 29.55$ $z = (29.55 - 29)/2.096 = 0.55/2.096 \approx 0.263$ Power = $P(Z < 0.263) \approx .6039 \rightarrow 60.39\%$ $\beta = 1 - .6039 \approx .3961$

Two-tailed power — do both tails

Score: $\mu_0 = 75$, $\sigma = 8$, $n = 16$, $\alpha = .05$, true mean 76 SE = $8/\sqrt{16} = 8/4 = 2$ cuts = $75 \pm 1.96(2) = 75 \pm 3.92 \rightarrow 71.08$ and 78.92 $z_{\text{low}} = (71.08 - 76)/2 = -2.46$ $z_{\text{high}} = (78.92 - 76)/2 = 1.46$ $\beta = .9279 - .0069 \approx .9209$ Power = $1 - .9209 \approx .0791$

Power is tiny because 76 is only half a standard error from 75 — a small gap is hard to detect.

	H ₀ IS TRUE	H ₀ IS FALSE
you reject	Type I = α	correct = Power
you fail to reject	correct	Type II = β

d Type I and Type II in context

SAY Type I: "concluding _____ when in reality _____." **Type II:** "concluding there is not enough evidence that _____ when in reality it is true."

e If an error was committed, which type?

SAY "We [**rejected / failed to reject**] H₀, so only a Type [**I / II**] error is possible: _____."

Set by your decision, not by the data.

Power from JMP

JMP DOE > Design Diagnostics > Sample Size and Power > One Sample Mean → fill Alpha, Std Dev, Difference to Detect, Sample Size, leave **Power** empty > Continue

Two-sided by default, so on a one-tailed question trust the 3 steps above. Leaving *Sample Size* empty instead answers "how big a sample do I need".

What moves power

Power rises with n↑, α ↑, a bigger gap $|\mu_{\text{true}} - \mu_0|$, and σ ↓. Raising n is the only one with no downside. Reduce Type I → lower α ; reduce Type II → raise power.

Sketch it

Two bell curves on one axis with a single vertical line at the cut-off through both. Top curve centred at μ_0 , tail past the cut-off shaded = α . Bottom curve centred at the true μ , the "don't reject" side shaded = β , the rest = **power**.

CONDITIONS — IDENTICAL FOR Q3 AND Q4. SAVE THE RESIDUALS FIRST.

FIRST ▼Response y > Save Columns > Residuals, then again for Studentized Residuals and Predicted Values

THEN **Linearity + constant variance:** ▼Response y > Row Diagnostics > Plot Residual by Predicted · **Normality:** Analyze > Distribution on the Residual column > ▼ > Normal Quantile Plot + Goodness of Fit

SAY **Linearity / constant variance:** shapeless band of even width around zero → met. **U or arch** → add a squared term. **Funnel** → spread not constant. **Normality:** dots follow the line, Shapiro-Wilk p = _____ > α. **Independence:** nothing to plot — argue from how the data were collected. **Outliers:** studentized residuals within ±3.

Q3 · SIMPLE REGRESSION & CORRELATION — one predictor

a Plot y against x. Comment. Is a line appropriate?

JMP Analyze > Fit Y by X > y into Y, Response, x into X, Factor > OK > ▼ > Fit Line

SAY "The relationship is [positive / negative], [linear / curved] and [weak / moderate / strong], with [no / one] obvious outlier. No curvature, so a straight line is appropriate — a quadratic or exponential model is not needed."

b Find the correlation r. Describe the strength.

JMP Analyze > Multivariate Methods > Multivariate > both columns into Y, Columns > OK > Correlations table, the off-diagonal cell

SAY "r = _____ — a [weak / moderate / strong] [positive / negative] linear relationship: _____ with higher x tend to have higher y."

|r| ≥ .8 strong · .5–.8 moderate · < .3 weak. Or r = ±√RSquare with the sign of the slope.

c Fit the regression. What is the least squares line?

JMP Analyze > Fit Model > y into Y, select x > Add > Personality Standard Least Squares > Run > Parameter Estimates, Estimate column

SAY Predicted _____ = _____ + _____ (_____)

Use Fit Model, not Fit Y by X — only Fit Model gives the CI columns and the saved intervals for parts h and i.

d Interpret the slope

JMP none — the Estimate on the x row

SAY "For each additional 1 _____, the predicted (average) _____ [increases / decreases] by _____."

Say predicted or average, and never claim x causes y. The intercept is the predicted y at x = 0 — add "but x = 0 is outside the data, so it has no practical meaning" when that is true.

e At the α level, is x a useful predictor?

JMP Fit Model report > Parameter Estimates > the x row > read t Ratio and Prob>|t|; the ANOVA Prob > F is the identical number in simple regression

SAY H₀: β₁ = 0 vs H₁: β₁ ≠ 0; t = b₁/SE(b₁) = _____, df = n–2 = _____, p = _____ → "Since p [$<$ / $>$] α, there [is / is not] enough evidence to say that _____ helps predict _____."

One-tailed? Halve Prob>|t| only if the sign of b₁ matches H₁. To test a slope against a number other than 0, use t = (b₁ – c)/SE, or see whether the CI contains c.

f Find R². What is the degrees of freedom?

JMP Summary of Fit for RSquare and RMSE; the DF column of the ANOVA table

SAY "R² = _____ — _____ % of the variation in _____ is explained by the linear relationship with _____. Error df = n–2 = _____, and RMSE = _____, so predictions are typically off by about that much."

g Confidence interval for the slope

JMP right-click inside Parameter Estimates > Columns > tick Lower 95% / Upper 95% · other levels first: ▼Response y > Set Alpha Level > 0.01 (99%) or Other → 0.02 (98%)

SAY b₁ ± t*·SE(b₁) = _____ ± _____ → "We are _____ % confident that each additional 1 _____ is associated with a change in the mean _____ of between _____ and _____."

Read it on the x row, not Intercept. Will it contain 0? Match the levels: 99% ↔ .01, 98% ↔ .02, 95% ↔ .05 — if p < α the interval must exclude 0.

h CI for the mean y, and a prediction interval for one case

JMP In the data table add a new row, type the x value, leave y blank → the row fills with the fit and both intervals. Then ▼Response y > Save Columns > Mean Confidence Interval (the average) and Indiv Confidence Interval (one new case)

SAY **Mean:** "We are _____ % confident the **average** _____ for all _____ with x = _____ is between _____ and _____." **Indiv:** "...that **one particular** _____ with x = _____ has y between _____ and _____."

"average / mean / for ALL units with x = ..." → **Mean.** "predict / a SINGLE / this one / a new ..." → **Indiv,** always wider. The blank y is what keeps the new row out of the fit.

THE FORMULAS BEHIND THE LINE

$\bar{x}$, $\bar{y}$ = the two averages · these three are never shown in JMP $S_{xx} = \Sigma(x-\bar{x})^2$ spread of the x's → slope, $SE(b_1)$ $S_{yy} = \Sigma(y-\bar{y})^2$ spread of the y's → r, R^2 $S_{xy} = \Sigma(x-\bar{x})(y-\bar{y})$ move together → slope, r $b_1 = S_{xy}/S_{xx}$ $b_0 = \bar{y} - b_1\bar{x}$ (b₁ first) $r = S_{xy}/\sqrt{(S_{xx}\cdot S_{yy})}$ $R^2 = r^2$ $e = y - \hat{y}$

Check your work: your b₀ and b₁ must match *Parameter Estimates*.

Q4 · MULTIPLE REGRESSION — same report, more rows

a Is y linearly related to the remaining predictors?

JMP Analyze > Multivariate Methods > Multivariate > put y AND every x into Y, Columns > OK > Correlations + Scatterplot Matrix; read the y row

SAY "Every plot in the y row is [positive / negative] and roughly linear with no curvature, so linear terms suit all of them. Strength differs: strong _____, weak _____."

b Which predictor is most highly correlated with y?

JMP same Correlations table — the largest |r| in the y row, ignoring y's own 1.0000

SAY " _____, r = _____ — the strongest linear association with y."

c Are any predictors highly correlated with each other?

JMP the x × x block of that same table · numerically: right-click inside Parameter Estimates > Columns > VIF

SAY "The strongest predictor–predictor pair is _____ (r = _____). All |r| < 0.8 and every VIF is under 5, so multicollinearity [is / is not] a problem and the coefficients stay interpretable."

VIF over 5 is a concern, over 10 serious. Name the pairs and give the numbers — don't just write "multicollinearity is present".

d Fit the model. Report it.

JMP Analyze > Fit Model > y into Y; click the first x then shift-click the last > Add > Personality Standard Least Squares, Emphasis **Minimal Report** > Run

SAY $\hat{Y} = \text{_____} + \text{_____} X_1 + \text{_____} X_2 + \text{_____} X_3 \dots$

e Interpret one coefficient

JMP the Estimate on that predictor's row

SAY "**Holding all the other predictors constant**, each additional 1 _____ is associated with a [rise / fall] of _____ in the mean _____."

That first clause is the entire difference from simple regression. Leave it out and you lose the mark.

f Find R², adjusted R², and the degrees of freedom

JMP Summary of Fit for RSquare, RSquare Adj and RMSE; the DF column of the ANOVA table

SAY "R² = _____ — _____ % of the variation in y is explained by the _____ predictors together. Error df = n–k–1 = _____."

R²_{adj} = 1 – (1–R²)(n–1)/(n–k–1). Plain R² always rises when a predictor is added, so only **adjusted** R² can compare models of different sizes.

g Two different tests — don't mix them up

SAY **Whole model:** ANOVA **F Ratio + Prob > F**, df = k and n–k–1. H₀: every slope is 0 → "the model [is / is not] useful for predicting y." Rejecting means **at least one** predictor helps. **One predictor, adjusting for the rest:** that row's **t Ratio + Prob>|t|**, df = n–k–1.

A predictor can look strong alone and still be useless here, because another one already covers it. CI for a coefficient: right-click *Parameter Estimates* > Columns > Lower/Upper 95%.

h How many models could you build? Which subset is best?

JMP Fit Model > add all > Personality **Stepwise** > Run > ▼ > **All Possible Models** → RSquare, RSquare Adj, Cp, AICc, BIC for every subset

SAY " $2^k - 1 = \text{_____}$ models — each predictor is either in or out, minus the empty model."

i Use backward selection to find the subset

JMP same Y and effects, Personality **Stepwise** > Run > set Stopping Rule = **P-value Threshold**, Direction = **Backward**, Prob to Leave = **0.1** > **Go** > finish with **Make Model** > Run

SAY "Drop the largest p above .10, one at a time: _____, then _____, then stop. The variables remaining are _____."

Current Estimates: a tick under **Entered** means it stayed. Remove one at a time — the p-values shift after every removal.

j Compare that model to the full one

JMP Summary of Fit for each: compare **RSquare Adj** and **RMSE**

SAY "Dropping _____ predictors costs only _____ points of R²; adjusted R² (_____ vs _____) and RMSE (_____ vs _____) are practically identical, so the smaller model **is preferred**."

"Best" = highest adjusted R² and lowest RMSE, never plain R².

k Predict from the model you chose

JMP add a row with those x values and a **blank y** > Run > Save Columns > Predicted Values, or Mean / Indiv Confidence Interval · or ▼Response y > Factor Profiling > Profiler

Use the coefficients of the model the question names, not the full model's.

+ Polynomial — when the scatterplot curves

JMP quick: Fit Y by X > ▼ > **Fit Polynomial** > 2, quadratic · full (needed for intervals): Fit Model > select x > **Macros** > **Polynomial to Degree**, Degree 2 > Run > adds an **x*x** row

SAY "The squared term has p = _____ [$<$ / $>$] α, so there [is / is not] enough evidence to say the curved model fits better than a straight line."

Test the **squared term only** — the overall F can look fine on the straight-line part alone. JMP **centres** the squared term by default and prints (x– $\bar{x}$)(x– $\bar{x}$); untick **Center Polynomials** on the ▼ to get the plain b₀, b₁, b₂.

The step in y is no longer constant

change = b₁(x₂ – x₁) + b₂(x₂² – x₁²) 0 → 1 is always b₁ + b₂ 2 → 3 is always b₁ + 5b₂ (because 3² – 2² = 5)

Put both x values in and subtract — never quote b₁ alone.